

VIDEO2SKILL: FROM STREAMING EXPERIENCE TO REUSABLE EMBODIED SKILLS

Jianshu Zhang^{1*} Ce Zhang^{2*} Xiyuan Yang³ Chenwei Xu¹ Haoran Lu¹ Yijiang Li⁴
 Yaqi Xie² Katia P. Sycara² Han Liu¹

¹Northwestern ²CMU ³UIUC ⁴UCSD *Equal contribution

Project Page: <https://andyzworks.github.io/video2skill>

ABSTRACT

Manipulation behaviors vary widely across objects and scenes, but they share a small set of reusable skills, and planning with these skills helps embodied agents generalize to new tasks. Yet an agent can only plan with skills it knows. Recovering skills from observed experience, the inverse of planning, builds this knowledge over time and yields skill data for training future agents. Vision-Language Models (VLMs) describe individual manipulation events well, but can they organize a stream of events into reusable skills? We formulate this problem as *Streaming Embodied Skill Discovery (SESD)*: a model watches videos in sequence and maintains a persistent skill library that shapes its later decisions. To systematically measure this ability, we introduce Video2Skill, a benchmark that covers robot tabletop manipulation and human kitchen activity and tests three core capabilities: (i) locating manipulation events in time, (ii) grouping events of the same transformation, and (iii) deciding when to reuse an existing skill or create a new one. Across 19 open-source VLMs, many models group events at near-chance level, and scale does not consistently help. Their errors depend on how perception and library updates are coupled: joint models merge distinct transformations into one skill, while models that update the library from text descriptions duplicate recurring ones. Supervised fine-tuning, including our counterfactual library-state rebalancing (CLARE), improves grouping but exposes a deeper bottleneck: trained models consolidate familiar skills yet rarely expand the library. Their libraries stall below half the reference size, and transformations unseen in training are located in time but almost never given a new skill. Recognizing when existing skills are insufficient thus emerges as the central challenge.

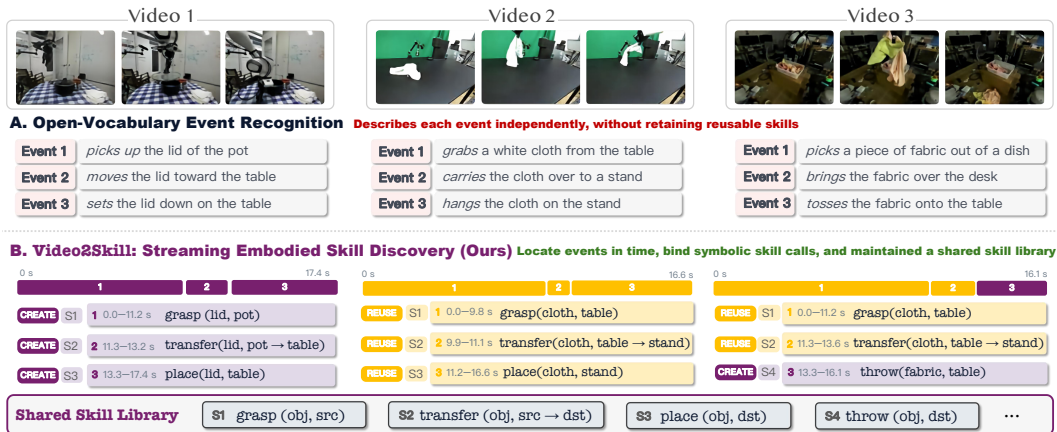


Figure 1: Video2Skill: From describing events to accumulating reusable skills. (A) Vision-Language Models recognize manipulation events in open-vocabulary language. (B) Video2Skill further organizes these observations into a persistent library of symbolic skills, grounding each event in time and deciding whether to reuse an existing skill or introduce a new one. Video 1 introduces grasp, transfer, and place; Video 2 reuses all three despite changes in objects and scene; Video 3 reuses grasp and transfer while adding throw for a new transformation. The library thus preserves shared structure across observations while expanding to accommodate novelty.

1 INTRODUCTION

Embodied agents that carry out long-horizon robotic tasks must plan. The behaviors they encounter are enormously diverse, varying in objects, scenes, and embodiments, yet they are underpinned by a much smaller set of shared skills: wiping a table, a plate, or a window all instantiate the same wiping skill. Planning over such skills therefore generalizes, as new tasks can be solved by recomposing familiar skills rather than learning each behavior from scratch (Ichter et al., 2023; Liang et al., 2023). Planning with skills, however, presupposes knowing which skills exist and how they are instantiated. A natural precursor is the inverse problem: observing continuous experience and recovering the skills that produced it (Baker et al., 2009). Much as people acquire skills by watching others and reuse them in new settings, an agent that solves this inverse problem can accumulate a skill library as experience unfolds, and the resulting skill-annotated experience can supervise future embodied agents (Kim et al., 2025; Xie et al., 2026). These skills are most useful when represented symbolically and with structure: a schema such as `grasp(object, source)` can be recognized across instances, composed by planners, and inspected by humans (Yang et al., 2025; Yu et al., 2026).

Vision-Language Models (VLMs) can already describe diverse manipulation events in open-vocabulary language (Bai et al., 2025; Chen et al., 2024; Zhang et al., 2025c; Pi et al., 2024). For agents operating over continuous visual streams (Qian et al., 2025; Wang et al., 2026a;c; Zhang et al., 2026a;d), a further challenge is to organize these events into a persistent skill library. Doing so requires locating events in time and deciding whether each instantiates a known skill or introduces a new one. Figure 1 illustrates this distinction: picking up a lid and grasping a cloth should reuse the same grasp schema with different arguments, whereas throwing fabric onto a table and placing it there involve distinct motion and release processes despite sharing a destination. These decisions become more challenging in a streaming setting, where the model relies on a library built from its own earlier predictions. Creating duplicate schemas fragments recurring experience, while reusing an overly broad schema merges distinct transformations. Because subsequent decisions depend on this evolving library, such errors can shape how later events are interpreted.

Existing approaches to skill reuse often rely on predefined repertoires or on signals beyond passive video observations. Some systems compose an existing set of robot skills (Ichter et al., 2023; Liang et al., 2023), while others grow their repertoires through code execution and environment feedback (Wang et al., 2024; Ning et al., 2026), task-success signals (Zhang et al., 2023), or robot trajectories (Wan et al., 2024). These settings differ from constructing a symbolic skill library online from a stream of visual observations. This leaves a fundamental question: *can a model build and maintain such a library from streaming video, using the abstractions established by its own earlier decisions to recognize familiar transformations and determine when a new schema is needed?*

We formulate this problem as *Streaming Embodied Skill Discovery (SESD)*. A model incrementally processes video, predicts the temporal extent of each manipulation event, and expresses it as a symbolic skill call: a reusable transformation schema bound to concrete arguments such as the object, source, and destination. The library persists across videos and conditions subsequent predictions, allowing schemas discovered in one scene to be reused in another. Skills here are semantic abstractions of observed transformations, rather than executable controllers. We introduce *Video2Skill*, a benchmark built from robot manipulation videos in ROBOINTER and egocentric kitchen recordings in HD-EPIC, to evaluate this capability across robotic and human activity. The benchmark evaluates how well models recover manipulation events, group instances of the same transformation regardless of schema names, and distinguish when to create or reuse a skill.

We evaluate 19 open-source VLMs using two processing paradigms: *unified* processing jointly recognizes events and updates the library, whereas *factorized* processing first generates textual event descriptions without library access and then uses the same model to update the library from those descriptions. Without task-specific training, many state-of-the-art models exhibit near-chance agreement with the reference grouping of events, revealing a substantial gap between event description and consistent skill abstraction. The two paradigms exhibit contrasting failure tendencies: unified models tend to merge distinct transformations into overly broad skills, while factorized models frequently create duplicate schemas for recurring transformations.

To investigate whether supervision can mitigate these failures, we fine-tune Qwen3.5-4B and LLaVA-OneVision-2-8B under both paradigms using three strategies: oracle-history supervised fine-tuning (SFT), Counterfactual Library-state Rebalancing (CLARE), and on-policy correction

using student-generated histories. Supervision improves skill grouping across both backbones, but models still struggle to distinguish new transformations from variations of familiar ones. The benefits of the three strategies vary across settings, highlighting the difficulty of learning when to preserve an existing skill and when to expand the library. Further analyses reveal where these difficulties persist. Training makes the schemas produced under different video orders more consistent, but its benefits are uneven across skills, with the most consistent gains on frequently demonstrated transformations. Transformations absent from task-specific training are often absorbed into existing schemas, even when their events are located in the video. These findings suggest that models can become better at organizing experience without reliably recognizing the limits of their existing repertoire. Developing models that both consolidate recurring transformations and introduce appropriate new abstractions remains a central challenge for learning from streaming visual experience.

Our contributions can be summarized as follows:

- We formulate *Streaming Embodied Skill Discovery* (SESD) and introduce `Video2Skill`, a benchmark built from ROBOINTER and HD-EPIC. Its evaluation jointly measures event coverage, naming-invariant skill grouping, and creation–reuse decisions across streaming videos.
- We evaluate 19 open-source VLMs under unified and factorized paradigms, identifying contrasting failure tendencies: unified models often merge distinct transformations, while factorized models frequently assign recurring transformations to duplicate schemas.
- We compare three supervision strategies across two backbones and both paradigms, finding improved grouping alongside persistent creation–reuse errors. Further analyses examine order consistency, uneven gains across training frequencies, and limited novelty recognition for transformations absent from task-specific supervision.

2 PROBLEM DEFINITION AND BENCHMARK CURATION

We introduce `Video2Skill` to study whether models can organize streaming visual experience into reusable skills. We first formalize Streaming Embodied Skill Discovery (SESD), then describe its construction from human and robot manipulation videos, and finally present an evaluation protocol covering event recovery, skill grouping, and creation–reuse decisions.

2.1 TASK FORMULATION

We process an ordered video stream $\mathcal{V} = (v_1, \dots, v_N)$ in chunks. At step t , the model receives chunk x_t and state $z_{t-1} = (\mathcal{L}_{t-1}, u_{t-1})$, containing a persistent skill library and an optional unfinished event:

$$y_t \sim \pi_\theta(\cdot \mid x_t, z_{t-1}), \quad y_t = (\mathcal{D}_t, \mathcal{E}_t, u_t). \quad (1)$$

Here, $\mathcal{D}_t$ contains new schemas, $\mathcal{E}_t$ contains completed events, and u_t carries an unfinished event across chunks. Each event $e = (b_e, d_e, s_e, a_e)$ specifies video-relative temporal boundaries, a schema, and argument bindings.

A schema $s = (n_s, d_s, \mathcal{A}_s)$ comprises a stable free-form name, a definition, and typed argument slots. It describes a physical transformation, not an executable controller. Reference equivalence is $e_i \sim e_j \iff \tau(e_i) = \tau(e_j)$, where $\tau(e)$ denotes the underlying transformation, independently of the model’s schema assignments.

Starting from $\mathcal{L}_0 = \emptyset$ and $u_0 = \emptyset$, the state updates as

$$\mathcal{L}_t = \mathcal{L}_{t-1} \cup \mathcal{D}_t, \quad z_t = (\mathcal{L}_t, u_t). \quad (2)$$

Events reference schemas in $\mathcal{L}_t$, either newly created or reused. At video boundaries, unfinished events are closed or discarded and u_t is reset; the library persists. Thus, earlier creation and reuse decisions shape the context for subsequent observations.

2.2 BENCHMARK CURATION

Source datasets. We use two public datasets spanning human and robot manipulation: HD-EPIC (Perrett et al., 2025), with long egocentric recordings of unscripted kitchen activity, and ROBOINTER, with robot tabletop videos drawn from DROID and RH20T (Khazatsky et al., 2024; Fang et al., 2024). The domains differ in action density and duration: median reference segments in the verified test splits last 1.9 s and 4.5 s, respectively.

Data curation. Our annotation process prioritizes visual evidence over preserving source labels. For HD-EPIC, we annotate selected recordings and clips throughout. For ROBOINTER, we check each candidate source segment against the video, correct annotation errors, and discard labels whose stated manipulation is not visually supported. Retained events are represented as typed skill calls, with argument bindings drawn from seven roles: object, source, destination, instrument, direction, substance, and state. Review corrected 241 of the 12,429 verified calls across the training and test splits. The retained ROBOINTER segments cover 55% of training video time and 82% of verified test video time, in contrast to the wall-to-wall native annotations. These references therefore represent verified manipulations rather than exhaustive temporal coverage. We organize verified calls into 48 canonical transformation classes with definitions and aliases, establishing the reference partition for naming-invariant evaluation. Models generate free-form schemas without receiving this class list.

Data splits. HD-EPIC provides 42 fully annotated training videos with 5,726 calls and 40 test clips from 29 held-out recordings with 1,137 reference segments. ROBOINTER provides 2,992 training videos with 5,032 verified calls and 132 test videos with 534 verified segments. The verified test splits contain 46 canonical classes in HD-EPIC and 15 in ROBOINTER. All training and test splits are disjoint at the source-video level, as verified programmatically.

2.3 EVALUATION PROTOCOL

We evaluate SESD along three complementary dimensions: event coverage, naming-invariant skill grouping, and creation-reuse decisions.

Temporal alignment and coverage. For video v , let $\mathcal{R}_v$ and $\mathcal{P}_v$ denote reference and predicted segments. Each reference independently selects the maximum-tIoU prediction, accepting matches at $\text{IoU} \geq 0.3$. Let $\mathcal{M}$ contain matched references across the episode:

$$p^*(r) = \arg \max_{p \in \mathcal{P}_v} \text{tIoU}(r, p), \quad \text{Coverage} = \frac{|\mathcal{M}|}{\sum_v |\mathcal{R}_v|}. \quad (3)$$

Predictions may match multiple references. Unmatched references enter only the coverage denominator; unmatched predictions incur no penalty. Coverage measures reference recovery.

Skill grouping. Index the $m = |\mathcal{M}|$ matched events in episode order, with reference classes c_i and aligned predicted names $\hat{s}_i$. Define

$$\begin{aligned} \mathcal{Q}_{\text{ref}} &= \{(i, j) : 1 \leq i < j \leq m, c_i = c_j\}, \\ \mathcal{Q}_{\text{pred}} &= \{(i, j) : 1 \leq i < j \leq m, \hat{s}_i = \hat{s}_j\}. \end{aligned} \quad (4)$$

Including cross-video pairs, we compute

$$P_{\text{pair}} = \frac{|\mathcal{Q}_{\text{ref}} \cap \mathcal{Q}_{\text{pred}}|}{|\mathcal{Q}_{\text{pred}}|}, \quad R_{\text{pair}} = \frac{|\mathcal{Q}_{\text{ref}} \cap \mathcal{Q}_{\text{pred}}|}{|\mathcal{Q}_{\text{ref}}|}. \quad (5)$$

Low precision indicates overmerging; low recall indicates fragmentation. Adjusted Rand Index (ARI) corrects partition agreement for chance under fixed cluster sizes. Grouping is invariant to consistent renaming, but synonymous strings remain distinct. Pair counts scale as $\binom{n}{2}$, favoring frequent classes; ARI does not ensure class balance.

Creation and reuse decisions. We traverse matched events in episode order and independently identify first occurrences of reference classes and predicted names:

$$g_i = \mathbf{1}[c_i \notin \{c_j : j < i\}], \quad \hat{g}_i = \mathbf{1}[\hat{s}_i \notin \{\hat{s}_j : j < i\}]. \quad (6)$$

One denotes CREATE and zero denotes REUSE. We report CREATE recall, $\Pr(\hat{g}_i = 1 \mid g_i = 1)$; REUSE recall, $\Pr(\hat{g}_i = 0 \mid g_i = 0)$; and decision accuracy, the fraction with $\hat{g}_i = g_i$. These measure novelty within the matched sequence, not explicit operations or complete library history, and exclude CONTINUE evaluation.

3 HOW DO CURRENT MODELS BEHAVE?

We evaluate 19 off-the-shelf VLMs on both domains using two inference paradigms (Table 1). *Unified* models jointly ground events and update the library from visual input and persistent state; *factorized* models first propose events without library access, then use the same model for text-only library

Table 1: **Raw model performance on SEDS**. Unified models jointly identify events and update the skill library; factorized models first propose events without library access, then use the same model for text-only library updates. Results use verified test splits in balanced order. Create R and Reuse R denote recall. All scores are multiplied by 100; ARI can be negative. Within each paradigm and column, first, second, and third ranks are highlighted.

Model	ROBOINTER						HD-EPIC					
	Cov.	Pair P	Pair R	ARI	Create R	Reuse R	Cov.	Pair P	Pair R	ARI	Create R	Reuse R
<i>Unified: a single VLM watches the video and manages the library in one pass</i>												
Qwen3.5-0.8B	2.4	36.4	77.4	-10.0	25.0	88.9	0.2	0.0	0.0	0.0	50.0	0.0
Qwen3.5-2B	39.3	31.1	72.8	8.0	7.7	99.5	14.8	9.2	53.8	5.1	9.1	99.3
Qwen3.5-4B	56.9	36.7	63.1	19.4	20.0	99.3	61.4	12.5	32.7	8.7	11.1	98.5
Qwen3.5-9B	44.8	24.9	34.2	1.7	21.4	99.1	53.9	15.5	26.1	11.8	20.0	98.2
Qwen3.5-27B	47.9	30.9	19.0	4.6	21.4	93.8	51.3	26.9	22.9	19.9	37.2	95.6
Qwen3.5-35B-A3B	57.1	29.6	37.1	4.3	13.3	99.0	55.9	19.7	19.9	13.9	29.5	96.1
Qwen3-VL-8B	56.0	30.0	59.1	12.4	20.0	100.0	44.1	13.1	48.9	11.5	12.2	97.2
Qwen3.8-27B	63.7	44.6	34.2	18.7	15.4	96.9	60.1	30.0	29.6	24.3	37.2	94.7
InternVL3.5-4B	41.6	36.0	72.1	12.6	15.4	99.5	19.0	7.6	54.7	2.3	16.2	98.3
InternVL3.5-8B	48.5	35.1	62.8	11.0	13.3	100.0	26.6	6.9	42.0	0.5	8.6	97.4
InternVL3.5-14B	50.9	28.8	43.3	6.1	16.7	98.5	38.0	12.5	33.6	9.0	17.5	96.9
InternVL3.5-38B	41.4	35.9	74.7	7.2	15.4	97.1	42.3	15.8	24.1	11.1	16.3	96.8
Ovis2.5-9B	4.9	24.7	45.1	-1.1	20.0	90.5	36.5	6.4	66.9	-0.5	4.5	98.9
Cosmos-Reason2-8B	48.5	43.2	80.3	27.0	6.7	99.6	37.6	8.7	58.2	4.4	20.5	98.2
Cosmos-Reason2-32B	54.3	32.5	34.6	2.9	21.4	96.7	44.4	12.0	31.0	8.5	24.4	92.5
LLaVA-OneVision-2-8B	37.8	22.5	100.0	0.0	7.1	100.0	21.6	13.4	52.4	13.2	12.8	95.7
GLM-4.1V-9B	40.1	28.1	76.8	5.1	7.1	97.5	5.8	4.2	21.2	-0.7	36.4	84.1
MiniCPM-V-4.5	43.8	25.4	58.6	7.5	26.7	99.5	7.5	24.7	60.6	25.6	26.1	93.5
Gemma-4-31B	52.6	36.6	36.6	12.5	7.1	95.9	46.0	15.5	25.6	12.0	34.9	94.0
<i>Factorized: the VLM first proposes events then updates the library via a text-only meta-policy</i>												
Qwen3.5-0.8B	26.0	0.0	0.0	-0.4	75.0	13.4	11.1	50.0	0.6	0.9	93.8	6.4
Qwen3.5-2B	42.3	23.1	0.2	-0.2	71.4	15.6	32.4	48.0	2.4	4.0	92.1	18.2
Qwen3.5-4B	46.6	31.3	8.9	3.2	26.7	68.8	50.1	6.3	15.9	-1.0	44.2	47.6
Qwen3.5-9B	50.2	24.3	10.1	0.5	46.7	70.8	49.2	21.4	4.3	5.0	62.8	41.5
Qwen3.5-27B	47.2	31.3	15.6	3.9	35.7	90.3	52.0	30.9	16.1	17.3	60.5	80.3
Qwen3.5-35B-A3B	57.3	23.1	0.4	-0.1	60.0	36.8	57.8	34.4	2.0	3.1	79.5	37.0
Qwen3-VL-8B	52.2	32.2	18.2	6.9	33.3	82.6	49.1	23.9	12.3	12.5	62.5	66.4
Qwen3.8-27B	43.5	28.9	16.7	0.8	33.3	90.3	52.7	29.7	22.3	20.8	40.9	87.6
InternVL3.5-4B	41.9	51.4	15.6	7.9	46.2	70.6	33.5	32.9	5.7	8.0	71.8	28.9
InternVL3.5-8B	38.4	53.7	10.5	5.5	50.0	51.8	34.7	25.3	3.3	4.4	76.9	27.8
InternVL3.5-14B	47.9	36.3	17.9	4.6	42.9	86.0	51.2	26.3	2.8	3.7	76.2	38.9
InternVL3.5-38B	48.3	44.9	7.5	4.1	61.5	62.9	49.7	27.7	5.1	6.5	67.4	45.8
Ovis2.5-9B	65.2	41.4	8.0	5.1	50.0	47.3	47.5	35.3	1.9	3.0	74.4	23.9
Cosmos-Reason2-8B	52.8	31.9	0.3	-0.0	64.3	26.5	50.7	21.3	4.9	5.6	63.6	44.0
Cosmos-Reason2-32B	47.2	40.6	4.1	1.7	53.8	67.4	28.8	17.5	6.5	6.2	71.0	45.3
LLaVA-OneVision-2-8B	71.7	27.4	0.8	0.1	80.0	29.1	40.5	17.2	0.6	0.7	87.8	16.0
GLM-4.1V-9B	36.5	17.4	0.1	-0.1	85.7	11.6	25.2	19.2	0.2	0.3	85.4	7.7
MiniCPM-V-4.5	67.4	30.1	0.2	0.1	80.0	14.8	43.0	26.3	0.2	0.4	88.1	9.2
Gemma-4-31B	54.9	28.4	21.6	5.8	35.7	93.9	51.9	22.5	15.3	13.8	50.0	75.5

updates. All evaluations use verified test splits, balanced video order, and no task-specific training. We examine pair precision and recall separately to distinguish overmerging from fragmentation.

Unified models tend to overmerge; factorized models tend to fragment. Unified models often reuse existing schemas even when a new transformation appears. Within the matched sequence, this tendency produces high reuse recall but low creation recall, while low pair precision confirms that distinct transformations are frequently grouped together. Factorized models create new schemas more readily, often improving pair precision. However, reuse and pair recall decline in nearly every comparison: the models become more willing to distinguish new transformations, but also repeatedly create separate schemas for transformations they have already encountered.

For example, on HD-EPIC, switching Qwen3.5-2B from unified to factorized raises pair precision from 9.2 to 48.0 but reduces pair recall from 53.8 to 2.4. Creation recall rises from 9.1 to 92.1, while reuse recall falls from 99.3 to 18.2. Higher creation recall thus accompanies a failure to reuse schemas for recurring transformations: the model separates different transformations more precisely, but also splits repeated instances across different names.

Scaling does not consistently improve skill abstraction. For unified Qwen3.5 on HD-EPIC, scaling from 4B to 27B increases pair precision from 12.5 to 26.9 but reduces pair recall from 32.7 to

22.9. Among matched events, the larger model exhibits less overmerging but greater fragmentation. ARI improves from 8.7 to 19.9, whereas the same scaling comparison on ROBOINTER reduces ARI from 19.4 to 4.6. InternVL3.5 likewise shows no monotonic improvement in ARI with size. Larger models can improve individual metrics, but do not consistently recover better skill partitions; ARI remains at or below 27.0 across all configurations.

Coverage and grouping quality must be interpreted together. Factorized LLaVA-OneVision-2-8B achieves the highest coverage on ROBOINTER (71.7%), yet its pair recall is only 0.8 and ARI is 0.1. Its unified counterpart exhibits the opposite problem: pair recall reaches 100.0, but pair precision is 22.5 and ARI is 0.0. Recovering more events therefore does not ensure consistent grouping, while high pair recall can result from grouping distinct transformations together. Partition scores also depend on which events are covered. Unified MiniCPM-V-4.5 achieves the highest ARI on HD-EPIC (25.6) while covering only 7.5% of reference events, so its grouping performance reflects a small matched subset. Similarly, unified Qwen3.5-0.8B attains creation recall of 50.0 with only 0.2% coverage, making that score uninformative about discovery across the full stream. Coverage, partition agreement, and creation–reuse decisions must therefore be interpreted jointly.

Implications. These results motivate two questions: whether models use the accumulated library to guide their decisions, and whether they correctly judge transformation equivalence across observations. The prevailing biases suggest complementary needs: recognizing novelty despite the availability of familiar schemas, and recognizing recurrence despite changes in objects, scenes, or event descriptions. Section 4 investigates how supervision under reference, counterfactual, and student-generated library histories affects these behaviors.

4 LEARNING TO MAINTAIN A SKILL LIBRARY

We investigate three supervised fine-tuning strategies that differ in the library states presented during training. Unified models learn to jointly ground events and update the library, whereas factorized models retain a frozen event proposer and train only the text-only meta-policy.

4.1 SUPERVISION STRATEGIES

The strategies share the supervised objective

$$\mathcal{L}_{\text{SFT}}(\theta) = -\mathbb{E}_{(o_t, z_{t-1}, y_t^*) \sim \mathcal{D}} \log p_\theta(y_t^* \mid o_t, z_{t-1}), \quad (7)$$

where o_t is a video chunk for unified models or a proposed event description for factorized models, and z_{t-1} contains the library and unfinished-event state. The target y_t^* specifies grounded events and library updates in the unified paradigm, or a library-management transaction in the factorized paradigm. Each strategy constructs a different training distribution $\mathcal{D}$.

Oracle-history SFT uses states obtained by replaying annotated episodes in order. Each observation is paired with the library and unfinished-event state produced by preceding reference outputs, and the target specifies the appropriate response under that history. The library persists across videos; unified training additionally uses periodic resets to provide repeated supervision for initialization. This strategy teaches correct decisions under reference histories.

Counterfactual Library-state Rebalancing (CLARE) augments these histories with edited states while holding the observation fixed. Removing a required schema changes reuse to creation, whereas adding semantically distinct lexical distractors or removing unrelated entries should preserve the correct assignment. The factorized variant also includes equivalent-schema insertion, which changes creation to reuse, and interventions on unfinished-event state. Targets are adjusted to the edited context. Counterfactual examples branch from the reference trajectory without modifying its subsequent states, supervising both sensitivity to relevant changes and consistency under irrelevant ones.

On-policy correction addresses the mismatch between reference histories and states encountered during inference. We roll out the student over ordered training episodes and obtain teacher-labeled targets for the states reached through its own predictions. These targets are recomputed for the encountered library: if the student failed to introduce a required schema earlier, a recurring transformation may now require creation even when the reference history prescribes reuse. Fine-tuning on these examples provides supervision under accumulated student errors. The objective remains supervised learning; student rollouts change the training-state distribution, similar to on-policy teacher–student distillation for language models (Agarwal et al., 2024; Gu et al., 2024).

Table 2: **Effect of supervision on Video2Skill across two backbones.** Pink highlights column maxima among the three supervised variants within each backbone and paradigm, including ties. All scores are multiplied by 100; ARI can be negative.

Method	ROBOINTER						HD-EPIC					
	Cov.	Pair P	Pair R	ARI	Create R	Reuse R	Cov.	Pair P	Pair R	ARI	Create R	Reuse R
<i>Unified: a single VLM watches the video and manages the library in one pass</i>												
Qwen3.5-4B	56.9	36.7	63.1	19.4	20.0	99.3	61.4	12.5	32.7	8.7	11.1	98.5
+ Oracle-history SFT	52.4	67.6	74.5	60.0	33.3	97.7	49.6	31.1	42.8	31.0	28.6	97.3
+ CLARE	59.2	66.1	70.1	56.2	20.0	97.0	53.3	31.8	37.7	29.2	40.0	97.5
+ On-policy correction	50.2	65.1	73.0	56.3	33.3	98.4	55.8	27.3	46.8	28.2	19.1	97.6
LLaVA-OneVision-2-8B	37.8	22.5	100.0	0.0	7.1	100.0	21.6	13.4	52.4	13.2	12.8	95.7
+ Oracle-history SFT	54.1	66.9	77.2	60.9	38.5	98.6	51.6	31.4	43.4	30.8	34.2	97.2
+ CLARE	60.9	62.4	78.3	58.1	33.3	98.7	55.3	26.4	42.3	25.9	35.0	97.5
+ On-policy correction	50.4	65.1	72.8	55.5	30.8	97.7	46.7	27.5	45.4	28.0	30.0	97.8
<i>Factorized: the VLM first proposes events then updates the library via a text-only meta-policy</i>												
Qwen3.5-4B	46.6	31.3	8.9	3.2	26.7	68.8	50.1	6.3	15.9	-1.0	44.2	47.6
+ Oracle-history SFT	46.6	33.8	27.4	10.4	20.0	93.2	50.1	26.7	22.9	19.5	39.5	90.5
+ CLARE	46.6	30.5	15.9	4.7	20.0	79.9	50.1	25.7	25.0	19.9	34.9	90.5
+ On-policy correction	46.6	33.9	31.4	11.4	20.0	94.0	50.1	26.0	23.8	19.5	37.2	93.5
LLaVA-OneVision-2-8B	71.7	27.4	0.8	0.1	80.0	29.1	40.5	17.2	0.6	0.7	87.8	16.0
+ Oracle-history SFT	71.7	30.9	10.1	2.7	46.7	74.7	40.5	18.4	5.5	5.7	61.0	60.1
+ CLARE	71.7	32.8	25.0	7.5	13.3	94.6	40.5	17.8	10.1	8.7	46.3	74.0
+ On-policy correction	71.7	30.8	25.1	5.5	20.0	94.6	40.5	16.8	13.9	10.1	46.3	84.2

4.2 EFFECTS OF SUPERVISION

We compare the three supervision strategies on Qwen3.5-4B and LLaVA-OneVision-2-8B under both paradigms and present the results in Table 2.

Supervision mitigates both failure tendencies. All three strategies improve ARI over their zero-shot counterparts across both backbones, domains, and paradigms. Unified models improve both pair precision and creation recall. For example, oracle-history SFT raises Qwen3.5-4B’s pair precision from 30.0 to 67.6 on ROBOINTER and from 12.7 to 31.1 on HD-EPIC, indicating less mixing of distinct transformations among matched events. These gains should be interpreted alongside changes in coverage: in the latter setting, coverage falls from 69.9 to 49.6, so the partition scores concern different matched subsets. Factorized models improve recurrence grouping while coverage remains fixed. For LLaVA-OneVision-2-8B on ROBOINTER, on-policy correction raises pair recall from 0.8 to 25.1 and reuse recall from 29.1 to 94.6, while creation recall falls from 80.0 to 20.0. Training therefore alleviates fragmentation, but also causes more first occurrences of reference transformations to receive previously used schema names.

Training-state interventions have different benefits across settings. Oracle-history SFT achieves the highest unified ARI in all four backbone–domain settings. Counterfactual and student-generated histories can improve other aspects of performance without improving overall partition agreement. On HD-EPIC, unified CLARE raises Qwen3.5-4B’s coverage from 49.6 to 53.3 and creation recall from 28.6 to 40.0 relative to oracle-history SFT, while ARI decreases from 31.0 to 29.2. The effects also depend on the backbone: in the factorized paradigm on ROBOINTER, CLARE improves LLaVA’s ARI from 2.7 to 7.5 but reduces Qwen’s from 10.4 to 4.7. On-policy correction attains the highest factorized reuse recall in every setting, including one tie, yet does not consistently achieve the highest ARI. The additional training-state interventions thus provide setting-specific benefits rather than uniform improvements over reference-history supervision.

Better grouping does not ensure correct creation and reuse. Despite improvements over zero-shot performance, trained unified models retain a substantial imbalance: reuse recall remains at 97.0–98.7, while creation recall reaches only 19.1–40.0. Most first occurrences of reference transformations in the matched sequence therefore still receive previously used names. Factorized models illustrate a complementary limitation: their improved reuse recall coexists with pair recall of only 5.5–31.4. Reuse recall records whether a schema name has appeared before, not whether it is the appropriate schema for the current transformation. A model can therefore reuse names frequently while selecting incorrect schemas or distributing recurring instances across duplicate entries. Supervision improves partition agreement, but reliable library maintenance still requires both recognizing novelty and preserving the identity of recurring transformations.

5 FURTHER DISCUSSIONS

How does video order affect skill discovery?

We investigate the effect of presentation order by replaying the same 40 HD-EPIC clips as closed-loop episodes. Alongside the default balanced order, we introduce a reuse-heavy order that spreads first occurrences of reference transformations throughout the episode and an innovation-heavy order that concentrates them near the beginning. Figure 2 shows the growth trajectories for oracle-history SFT. Despite different trajectories, the libraries end with similar sizes of 21–22 schemas, compared with 46 reference classes. Under the innovation-heavy order, all 46 reference classes have appeared by the twentieth video, yet the model has introduced only 20 schemas. Under the other orders, reference classes continue to accumulate later while the predicted libraries expand slowly. Making novelty appear earlier therefore does not substantially enlarge the model’s final repertoire. Beyond library size, oracle-history SFT achieves a mean pairwise Jaccard similarity of 0.78 between final schema-name sets across orders, compared with 0.59 for the zero-shot model; CLARE also achieves a mean of 0.78. Training thus improves consistency in the names retained across orders, while limited library expansion persists.

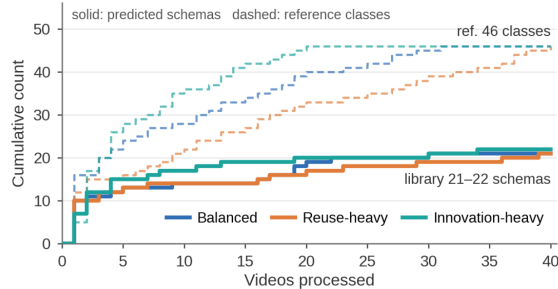


Figure 2: **Library growth under different video orders.** Unified Qwen3.5-4B with oracle-history SFT on the 40 HD-EPIC clips. Solid and dashed curves show cumulative predicted schemas and observed reference classes, respectively. All three orders end with 21–22 schemas versus 46 reference classes.

Are training gains shared across skills? We group training-seen reference classes by their frequency in the training annotations and report class-averaged pair F1 (Table 3a). High-frequency classes benefit consistently from supervision: their scores rise from 19.0 to 34.9–40.0 on HD-EPIC and from 25.5 to 40.9–51.7 on ROBOINTER. Improvements elsewhere are smaller and depend on the training strategy. Medium-frequency classes remain below 10 on both datasets, while low-frequency classes improve under oracle-history SFT and on-policy correction but not consistently under CLARE. Performance also does not vary monotonically with frequency: low-frequency groups sometimes outperform medium-frequency groups, including under oracle-history SFT and on-policy correction on both datasets. The results therefore support uneven benefits across the skill repertoire, with the most consistent gains among frequently demonstrated transformations, but do not establish a simple relationship between training frequency and grouping quality.

Do models recognize unseen transformations as novel? Eleven HD-EPIC test classes, comprising 59 segments, are absent from task-specific training annotations. Their temporal coverage after supervision is 57.6–66.1, exceeding the corresponding seen-class coverage under every strategy, as presented in Table 3b. However, this recovery rarely leads to creation: trained models assign a previously unused schema name at the first matched occurrence of only zero or one of the nine to ten covered unseen classes. Seen-class creation recall, in contrast, improves from 14.7 to 34.4–48.4. Higher grouping scores do not resolve this gap: oracle-history SFT raises unseen-class macro pair F1 from 8.6 to 25.5, yet creates a new name at the first matched occurrence of only one of ten covered classes. Improved grouping therefore does not establish novelty recognition, nor does failure to create at the first matched occurrence establish that a schema is never introduced later. These results highlight the difficulty of recognizing when existing schemas are insufficient, even among events that are successfully matched in time.

6 RELATED WORK

Video understanding and streaming models. Action recognition, temporal localization, and procedural understanding typically evaluate predictions against predefined action or keystep categories (Damen et al., 2022; Grauman et al., 2022; 2024; Perrett et al., 2025; Sener et al., 2022; Zhukov et al., 2019; Zhang et al., 2026e,f,g; Liu et al., 2026), while video QA and long-video benchmarks assess responses to supplied questions (Xiao et al., 2021; Mangalam et al., 2023; Fu et al., 2025; Wu et al., 2024; Zhang et al., 2025b). Unsupervised action segmentation goes beyond predefined categories by discovering recurring action structure, typically through offline clustering

Table 3: **Skill grouping and novelty recognition with unified Qwen3.5-4B.** (a) Macro pair F1 by training frequency (class counts in parentheses). (b) Classes seen or unseen in task-specific training. For unseen classes, k/n counts those receiving a new schema name at their first matched occurrence, out of all covered classes. Other scores are multiplied by 100.

Frequency	Zero-shot	Oracle history	CLARE	On-policy correction	Metric	Zero-shot	Oracle history	CLARE	On-policy correction
HD-EPIC					Training-seen				
High (11)	19.0	38.3	34.9	40.0	Coverage	61.3	48.7	53.1	51.8
Mid (13)	4.2	7.5	9.7	7.0	Create R	14.7	34.4	48.4	34.4
Low (11)	8.5	15.4	7.9	17.3	Training-unseen				
ROBOINTER					Coverage	62.7	66.1	57.6	66.1
High (5)	25.5	45.0	40.9	51.7	Macro pair F1	8.6	25.5	23.2	10.2
Mid (5)	2.5	1.5	1.7	2.9	Create (k/n)	0/11	1/10	1/9	0/10
Low (5)	3.8	5.8	5.1	7.1					

(a) Grouping by training frequency

(b) Seen vs. unseen on HD-EPIC

over complete sequences (Sener & Yao, 2018; Kukleva et al., 2019). Online action detection and streaming video models support incremental processing and temporal memory (Xu et al., 2019; 2021; Zhang et al., 2025a; Song et al., 2024; He et al., 2024; Zhang et al., 2026a), with dedicated benchmarks evaluating understanding as observations arrive (Lin et al., 2026; Niu et al., 2025). Video2Skill focuses on how recurring events are consolidated into reusable transformation schemas. Models assign temporally grounded events to a library, deciding when to reuse an existing schema or introduce a new one, with the resulting library conditioning subsequent predictions.

Skill discovery and evolving concept inventories. Reinforcement learning studies temporally extended behaviors acquired through interaction (Sutton et al., 1999; Bacon et al., 2017; Eysenbach et al., 2019; Sharma et al., 2020; Park et al., 2024), while imitation and continual learning infer skills or expand behavioral repertoires from demonstrations (Pertsch et al., 2021; Ajay et al., 2021; Zhu et al., 2022; Wan et al., 2024; Zhang et al., 2023; 2026b). XSkill (Xu et al., 2023) and UniSkill (Kim et al., 2025) learn shared skill representations from human and robot videos for downstream control; XSkill uses a fixed-size set of learned prototypes. LLM agents can grow explicit skill libraries online, typically using execution and environment feedback to validate new skills (Wang et al., 2024; 2025a;b; Zheng et al., 2025; Luo et al., 2026), and research agents can evolve reusable procedures through proactive online exploration (Wang et al., 2026b). Related approaches develop symbolic models for composing robot skills, as in SkillWrapper (Yang et al., 2025), or build video-derived repositories that support planning and skill expansion, as in Uni-Skill (Xie et al., 2026). The decision to expand a library also connects to novel and generalized category discovery (Han et al., 2019; Vaze et al., 2022) and their continual variants (Zhang et al., 2022; Wu et al., 2023; Ma et al., 2024), as well as to continual learning (Zhang et al., 2024; Rong et al., 2025) and in-context learning (Brown et al., 2020; Pi et al., 2025): creating a schema accommodates a new concept, while reusing one exploits concepts already held in context. Video2Skill studies this familiar-versus-novel distinction for temporally grounded transformations with explicit argument slots, allowing instances involving different objects and scenes to share a schema.

7 CONCLUSION

We introduced Video2Skill to study whether models can turn streaming visual experience into a persistent library of reusable skills. Across robot manipulation and egocentric kitchen activity, evaluations of 19 VLMs reveal complementary weaknesses: unified models tend to merge distinct transformations, while factorized models frequently assign recurring transformations to duplicate schemas. Oracle-history SFT, counterfactual library-state rebalancing, and on-policy correction partially alleviate these failures, but do not reliably resolve the boundary between novelty and reuse. Our findings underscore the need to evaluate event coverage and abstraction quality jointly, and establish a testbed for learning coherent skill libraries that remain useful as experience accumulates.

Limitations. Video2Skill focuses on robot manipulation and egocentric kitchen activity. Although these domains provide varied objects and transformations, broader settings such as outdoor activities remain unexplored. Our skill schemas describe observed physical transformations; their utility for downstream planning and executable control requires further evaluation.

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